

baseMean	log2Fold	Change	lfcSE	stat	pvalue	padj	Kingdom	Phylum	Class	Order	Family	Genus	Species	combined_traits
52.44158	7.5719	1.04291	7.26034	3.86E-13	4.49E-10	k_Bacteria	p_Verrucomicrobiota	c_Verrucomicrobiae	o_Pedospirales	f_Pedospiraceae	g_Ellin516	s_	unassigned	
55.98789	6.17913	0.90597	6.8205	9.07E-12	7.03E-09	k_Bacteria	p_Proteobacteria	c_Gammaproteobacteria	o_Pseudomonadales	f_Pseudomonadaceae	g_Pseudomonas	s_	aerobic_chemoheterotrophy	
68.58322	5.93009	0.96128	6.16894	6.87E-10	2.67E-07	k_Bacteria	p_Acidobacteriota	c_Acidobacteriae	o_Acidomondales	f_	g_	s_	unassigned	
32.46173	5.60231	0.95701	5.85398	4.80E-09	1.40E-06	k_Bacteria	p_Bacteroidia	c_Bacteroidia	o_Flavobacteriales	f_Blattabacteriaceae	g_Candidatus Brownia	s_	unassigned	
39.24658	5.29879	1.01671	5.21172	1.87E-07	3.96E-05	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Frankiales	f_Acidothermaceae	g_Acidothermus	s_	cellulolysisaerobic_chemoheterotrophy	
32.62675	4.94216	1.02998	4.7983	1.60E-06	2.49E-04	k_Bacteria	p_Bacteroidia	c_Bacteroidia	o_Sphingobacteriales	f_Sphingobacteriaceae	g_Mucilaginibacter	s_	unassigned	
15.28378	4.85335	1.04161	4.65946	3.17E-06	4.10E-04	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Frankiales	f_Acidothermaceae	g_Acidothermus	s_	cellulolysisaerobic_chemoheterotrophy	
23.97129	4.76844	0.80562	5.88074	3.87E-09	1.29E-06	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Frankiales	f_Acidothermaceae	g_Acidothermus	s_	cellulolysisaerobic_chemoheterotrophy	
49.30368	4.7482	0.85777	5.3355	3.10E-08	8.02E-06	k_Bacteria	p_Anarolineae	c_Anarolineae	o_	f_	g_SBR1331	s_	unassigned	
71.4423	4.63257	0.96719	4.79077	1.66E-06	2.49E-04	k_Bacteria	p_Proteobacteria	c_Alphaproteobacteria	o_Elterales	f_URHD0088	g_	s_	aerobic_chemoheterotrophy	
17.71275	4.54921	1.04019	4.37343	1.22E-05	1.24E-03	k_Bacteria	p_Acidobacteriota	c_Acidobacteriae	o_Solibacterales	f_Solibacteraceae	g_Candidatus Solibacter	s_	unassigned	
102.454	4.53255	0.67358	6.72902	1.71E-11	9.93E-09	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Frankiales	f_Acidothermaceae	g_Acidothermus	s_	cellulolysisaerobic_chemoheterotrophy	
15.09497	4.51515	1.00814	4.4787	7.51E-06	9.19E-04	k_Bacteria	p_Proteobacteria	c_Gammaproteobacteria	o_KF-JG30-C25	f_	g_	s_	unassigned	
17.96704	4.51218	1.01111	4.46258	8.10E-06	9.42E-04	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Frankiales	f_Acidothermaceae	g_Acidothermus	s_	cellulolysisaerobic_chemoheterotrophy	
21.18937	4.21621	0.78182	5.39279	6.94E-08	1.61E-05	k_Bacteria	p_Verrucomicrobiota	c_Verrucomicrobiae	o_Pedospirales	f_Pedospiraceae	g_	s_	unassigned	
15.58998	4.16464	0.84752	4.91389	8.93E-07	1.73E-04	k_Bacteria	p_Myxococcota	c_Polyangia	o_Polyangiales	f_Phaseolicystidaceae	g_Phaseolicystis	s_	unassigned	
72.82984	3.98018	0.82371	4.83201	1.35E-06	2.42E-04	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Catenulisporales	f_Catenulisporaceae	g_Catenulispora	s_	unassigned	
12.29496	3.95457	0.95065	4.15984	3.18E-05	2.55E-03	k_Bacteria	p_Acidobacteriota	c_Acidobacteriae	o_	f_	g_	s_	unassigned	
9.66996	3.80005	0.96958	3.91929	8.88E-05	5.44E-03	k_Bacteria	p_Chloroflexi	c_Anarolineae	o_	f_	g_	s_	unassigned	
13.56281	3.75285	0.90996	4.12421	3.72E-05	2.74E-03	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Isoisphaerales	f_Isoisphaeraeae	g_Singulisphaera	s_	aerobic_chemoheterotrophyureolysis	
33.92998	3.68173	0.84363	4.36416	1.28E-05	1.24E-03	k_Bacteria	p_Plantomycetota	c_Physcisphaerae	o_Tepidispisphaerales	f_WD2101 soil group	g_	s_	unassigned	
12.32559	3.65173	0.85672	4.26247	2.02E-05	1.81E-03	k_Bacteria	p_Verrucomicrobiota	c_Verrucomicrobiae	o_Chthoniobacterales	f_Chthoniobacteraceae	g_Chthoniobacter	s_	aerobic_chemoheterotrophy	
18.65411	3.6331	0.75927	4.78499	1.71E-06	2.49E-04	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Gemmatales	f_Gemmataceae	g_	s_	unassigned	
40.05035	3.56413	0.86035	4.14263	3.43E-05	2.66E-03	k_Bacteria	p_Actinobacteriota	c_Acidimicrobia	o_	f_	g_	s_	unassigned	
10.74846	3.55052	0.84963	4.17888	2.93E-05	2.43E-03	k_Bacteria	p_Plantomycetota	c_Physcisphaerae	o_Tepidispisphaerales	f_WD2101 soil group	g_	s_	aerobic_chemoheterotrophy	
9.19195	3.50125	0.93036	3.76333	1.68E-04	9.29E-03	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Plantomycetales	f_Schlesneriaceae	g_Schlesneria	s_	on	
8.52027	3.4599	0.93234	3.71099	2.06E-04	1.09E-02	k_Bacteria	p_Plantomycetota	c_Physcisphaerae	o_Tepidispisphaerales	f_WD2101 soil group	g_	s_	unassigned	
8.65025	3.45387	0.89061	3.87811	1.05E-04	6.28E-03	k_Bacteria	p_Patescibacteria	c_Saccharimonadia	o_Saccharimonadales	f_	g_	s_	unassigned	
301.06511	3.40161	0.52055	6.53466	6.38E-11	2.97E-08	k_Bacteria	p_Chloroflexi	c_Ktedonobacteria	o_Ktedonobacterales	f_Ktedonobacteraceae	g_JG30a-KF-32	s_	unassigned	
12.71278	3.3962	0.93011	3.65138	2.61E-04	1.26E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Gemmatales	f_Gemmataceae	g_	s_	unassigned	
33.58981	3.30281	1.03302	3.19725	1.39E-03	3.47E-02	k_Bacteria	p_Chloroflexi	c_Ktedonobacteria	o_Ktedonobacterales	f_Ktedonobacteraceae	g_1959-1	s_	unassigned	
5.3701	3.25948	0.81227	4.0128	6.00E-05	4.23E-03	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Frankiales	f_Acidothermaceae	g_Acidothermus	s_	cellulolysisaerobic_chemoheterotrophy	
4.515927	3.25749	1.04318	3.12266	1.01E-06	4.01E-02	k_Bacteria	p_Chloroflexi	c_Ktedonobacteria	o_Ktedonobacterales	f_JG30-KF-AS9	g_	s_	unassigned	
5.38146	3.25712	0.87395	3.72691	1.94E-04	1.05E-02	k_Bacteria	p_Myxococcota	c_Polyangia	o_Polyangiales	f_Polyangiaceae	g_Sorangium	s_	cellulolysisaerobic_chemoheterotrophy	
8.02639	3.22947	0.93363	3.45904	5.42E-04	2.10E-02	k_Bacteria	p_Acidobacteriota	c_Blastocatella	o_11-24	f_	g_	s_	unassigned	
25.15908	3.16467	0.71417	4.43125	9.37E-06	1.01E-03	k_Bacteria	p_Actinobacteriota	c_Thermoleophilina	o_Solirubrobacterales	f_Solirubrobacteraceae	g_Conexibacter	s_	unassigned	
4.42085	3.10526	0.97861	3.17314	1.51E-03	3.65E-02	k_Bacteria	p_Plantomycetota	c_Physcisphaerae	o_Tepidispisphaerales	f_WD2101 soil group	g_	s_	unassigned	
7.25141	3.08006	0.9079	3.39522	6.93E-04	2.40E-02	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Frankiales	f_Acidothermaceae	g_Acidothermus	s_	cellulolysisaerobic_chemoheterotrophy	
5.059	3.07554	0.94037	3.27114	1.07E-03	3.14E-02	k_Bacteria	p_Proteobacteria	c_Alphaproteobacteria	o_	f_	g_	s_	unassigned	
8.64914	3.07131	0.88785	3.45927	5.42E-04	2.10E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Isoisphaerales	f_Isoisphaeraeae	g_Tundrisphaera	s_	aerobic_chemoheterotrophy	
30.8692	3.06279	0.85672	3.575	3.50E-04	1.54E-02	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Frankiales	f_	g_	s_	unassigned	
6.87523	2.97839	0.83458	3.56873	3.59E-04	1.55E-02	k_Bacteria	p_Verrucomicrobiota	c_Verrucomicrobiae	o_Chthoniobacterales	f_Terrimicrobiaceae	g_Terrimicrobium	s_	unassigned	
31.04923	2.95774	0.69567	4.25162	2.12E-05	1.83E-03	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Gemmatales	f_Gemmataceae	g_	s_	unassigned	
28.88157	2.94657	0.84657	3.48059	5.00E-04	2.01E-02	k_Bacteria	p_Actinobacteriota	c_Acidimicrobia	o_	f_	g_	s_	unassigned	
4.84388	2.94245	0.91542	3.21431	1.31E-03	3.36E-02	k_Bacteria	p_Armatimonadota	c_Fimbrimonadia	o_Fimbrimonadales	f_Fimbrimonadaceae	g_Fimbrimonas	s_	unassigned	
11.92956	2.92017	0.87789	3.32637	8.80E-04	2.77E-02	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Frankiales	f_Acidothermaceae	g_Acidothermus	s_	cellulolysisaerobic_chemoheterotrophy	
3.88605	2.9055	0.94399	3.07788	2.08E-03	4.41E-02	k_Bacteria	p_Chloroflexi	c_AD3	o_	f_	g_	s_	unassigned	
20.54711	2.89957	0.76262	3.80213	1.43E-04	8.14E-03	k_Bacteria	p_Bacteroidia	c_Bacteroidia	o_Chitinophagales	f_Chitinophagaceae	g_	s_	unassigned	
5.05227	2.8964	0.9065	3.19513	1.40E-03	3.47E-02	k_Bacteria	p_Verrucomicrobiota	c_Verrucomicrobiae	o_Chthoniobacterales	f_Terrimicrobiaceae	g_Terrimicrobium	s_	unassigned	
13.36268	2.86218	0.81323	3.44329	5.75E-04	2.14E-02	k_Bacteria	p_Armatimonadota	c_Fimbrimonadia	o_Fimbrimonadales	f_Fimbrimonadaceae	g_	s_	unassigned	
3.86507	2.84878	0.91407	3.1166	1.83E-03	4.05E-02	k_Bacteria	p_Gemmatimonadota	c_Gemmatimonadates	o_Gemmatimonadales	f_Gemmatimonadaceae	g_	s_	unassigned	
8.7726	2.82956	0.63916	4.427	9.56E-06	1.01E-03	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Pirellulales	f_Pirellulaceae	g_Pirellula	s_	unassigned	
6.89939	2.82027	0.82447	3.4207	6.25E-04	2.24E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Isoisphaerales	f_Isoisphaeraeae	g_	s_	aerobic_chemoheterotrophy	
39.25115	2.79877	0.83369	3.35708	7.88E-04	2.59E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Gemmatales	f_Gemmataceae	g_	s_	unassigned	
9.47247	2.76579	0.8847	3.23596	1.21E-03	3.24E-02	k_Bacteria	p_Bacteroidia	c_Bacteroidia	o_Sphingobacteriales	f_Sphingobacteriaceae	g_Mucilaginibacter	s_	unassigned	
18.10066	2.76316	0.71952	3.84028	1.23E-04	7.15E-03	k_Bacteria	p_Proteobacteria	c_Alphaproteobacteria	o_Microspales	f_Microspagaceae	g_	s_	unassigned	
4.78469	2.76311	0.87842	3.14557	1.66E-03	3.87E-02	k_Bacteria	p_Myxococcota	c_Polyangia	o_Polyangiales	f_Polyangiaceae	g_	s_	unassigned	
10.78618	2.73476	0.69557	3.93167	8.44E-05	5.30E-03	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Gemmatales	f_Gemmataceae	g_Gemmata	s_	unassigned	
16.64555	2.70258	0.56858	4.75319	2.00E-06	2.74E-04	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Gemmatales	f_Gemmataceae	g_	s_	unassigned	
5.14327	2.69573	0.8573	3.14443	1.66E-03	3.87E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Gemmatales	f_Gemmataceae	g_	s_	unassigned	
37.34174	2.68424	0.78054	3.43896	5.84E-04	2.14E-02	k_Bacteria	p_Proteobacteria	c_Gammaproteobacteria	o_Burkholderiales	f_Comamonadaceae	g_Sphaerotilus	s_	aerobic_chemoheterotrophy	
13.15401	2.67778	0.87867	3.04755	2.31E-03	4.79E-02	k_Bacteria	p_Proteobacteria	c_Gammaproteobacteria	o_	f_	g_Acidibacter	s_	unassigned	
5.86568	2.61634	0.77792	3.36324	7.70E-04	2.59E-02	k_Bacteria	p_Proteobacteria	c_Alphaproteobacteria	o_Azospirillales	f_Inquilineae	g_Inquillus	s_	unassigned	
10.38918	2.60039	0.80662	3.2238	1.27E-03	3.34E-02	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Frankiales	f_Acidothermaceae	g_Acidothermus	s_	cellulolysisaerobic_chemoheterotrophy	
5.35734	2.57425	0.84191	3.05764	2.23E-03	4.67E-02	k_Bacteria	p_Plantomycetota	c_Physcisphaerae	o_Tepidispisphaerales	f_CPla-3 termite group	g_	s_	unassigned	
4.59724	2.57213	0.71092	3.61804	2.97E-04	1.35E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Gemmatales	f_Gemmataceae	g_	s_	unassigned	
15.97901	2.56636	0.78353	3.26779	1.08E-03	3.14E-02	k_Bacteria	p_Armatimonadota	c_Chthonomonadetes	o_Chthonomonadales	f_Chthonomonadaceae	g_Chthonomonas	s_	unassigned	
44.96587	2.53771	0.75931	3.34213	8.31E-04	2.66E-02	k_Bacteria	p_Verrucomicrobiota	c_Verrucomicrobiae	o_Pedospirales	f_Pedospiraceae	g_ADurb.Bin063-1	s_	unassigned	
2.65133	2.47526	0.75087	3.29653	9.79E-04	2.96E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Gemmatales	f_Gemmataceae	g_Gemmata	s_	unassigned	
8.94232	2.46248	0.78635	3.13154	1.74E-03	3.93E-02	k_Bacteria	p_Acidobacteriota	c_Acidobacteriae	o_Acidobacterales	f_	g_	s_	unassigned	
4.31177	2.44179	0.75002	3.25564	1.13E-03	3.15E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Gemmatales	f_Gemmataceae	g_	s_	unassigned	
4.96159	2.41804	0.66532	3.6344	2.79E-04	3.24E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Isoisphaerales	f_Isoisphaeraeae	g_Tundrisphaera	s_	aerobic_chemoheterotrophy	
11.75312	2.39849	0.88023	3.31588	9.14E-04	2.80E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Gemmatales	f_Gemmataceae	g_	s_	unassigned	
18.9078	2.37528	0.72777	3.2641	1.10E-03	3.14E-02	k_Bacteria	p_Acidobacteriota	c_Acidobacteriae	o_Acidobacterales	f_	g_	s_	unassigned	
6.67178	2.34435	0.76031	3.08177	2.06E-03	4.39E-02	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Frankiales	f_Acidothermaceae	g_Acidothermus	s_	cellulolysisaerobic_chemoheterotrophy	
2.43545	2.34332	0.74634	3.13976	1.69E-03	3.89E-02	k_Bacteria	p_Acidobacteriota	c_Acidobacteriae	o_Bryobacterales	f_Bryobacteraceae	g_Bryobacter	s_	aerobic_chemoheterotrophy	
6.07783	2.34262	0.71821	3.26175	1.11E-03	3.14E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Isoisphaerales	f_Isoisphaeraeae	g_Tundrisphaera	s_	aerobic_chemoheterotrophy	
5.60857	2.3362	0.72645	3.21592	1.30E-03	3.36E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Gemmatales	f_Gemmataceae	g_	s_	unassigned	
4.59721	2.29137	0.6982	3.28181	1.03E-03	3.08E-02	k_Bacteria	p_Actinobacteriota	c_Actinobacteria	o_Frankiales	f_Acidothermaceae	g_Acidothermus	s_	cellulolysisaerobic_chemoheterotrophy	
39.33923	2.27894	0.62171	3.66558	2.47E-04	1.22E-02	k_Bacteria	p_Plantomycetota	c_Plantomycetes	o_Isoisphaerales	f_Isoisphaeraeae	g_	s_	aerobic_chemoheterotrophy	
10.99739	2.21517	0.6285	3.52453	4.24E-04	1.76E-02	k_Bacteria	p_Acidobacteriota	c_Acidobacteriae	o					

23.14668	-30	0.83077	-36.11103	1.52E-285	3.54E-282	k__Bacteria	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__Lautropia	s__	fermentationaerobic_chemoheterotroph y
----------	-----	---------	-----------	-----------	-----------	-------------	-------------------	------------------------	--------------------	---------------------	--------------	-----	---